

Digital Wave Mechanics

*From Newton's Prism to Standing Waves
in Finite Golden Fields*

A companion work bridging *Logical Creativity* (La Rue) and *Physical Creativity* (Lockwood)

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Formal verification: [ProtorealZeta](#)

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Rigorous Expanded Edition

Abstract

We trace spectral decomposition from Newton's prism through Schrödinger's wave mechanics and Feynman's quantum chromodynamics to its discrete arithmetic completion in finite golden fields. The central algebraic result is precise: at the golden split primes $p = 181$ and $p = 229$, selected golden roots generate finite cyclic subgroups whose orders contain the factors 45, 57, 90, and 114, and the conjugate orbit at $p = 229$ contains a canonical order-three subgroup generated by $\omega = 94$ with $1 + \omega + \omega^2 \equiv 0 \pmod{229}$. This order-three subgroup is isomorphic to the center $Z(\mathrm{SU}(3)) \cong \mathbb{Z}/3\mathbb{Z}$. The paper therefore establishes a finite-field realization of the center algebra of color grading, not a proof of full dynamical QCD confinement. A survey of all 45 golden split primes below 500 confirms the confinement identity holds in all 16 cases where $3 \mid \mathrm{ord}(\bar{\varphi})$.

Walker's infoton, the minimum quantum of information-energy at the Landauer bound, is modeled structurally by the anchor element $\iota = P(0, 0, 1, 0, 0)$ of the Proto-real manifold: it has null bridge norm, period-two oscillation under the specified Klein product, and a three-arc grading inherited from the conjugate finite-field orbit. These correspondences define digital wave mechanics: interference, standing modes, spectral decomposition, center grading, and information-cost bookkeeping realized inside finite multiplicative groups and finite cyclic transforms, with physical interpretation separated from algebraic proof. The Prime-Dimensional Simplex $\Delta_p^4 = \{1, \omega, \omega^2, -1, \varphi\}$ provides the categorical scaffold: its five vertices correspond one-to-one with the five Cyber Prismatic formulas, and its tetrahedron face $\{1, \omega, \omega^2, -1\}$ recovers the Golden Tetrahedron of Golden Chromodynamics [10]. As a companion work between La Rue's *Logical Creativity* [11] and Lockwood's *Physical Creativity* [6], this paper provides the spectral bridge where abstract algebraic structure meets physical observation.

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Expansion Protocol and Scope. This document uses three claim statuses. A *theorem* is an algebraic or analytic statement proved from stated hypotheses. A *verified computation* is a finite calculation whose inputs and outputs are given explicitly and whose replication requires only modular exponentiation or the displayed script. A *physical interpretation* is a map from the algebraic object to physical language. Physical interpretations are not promoted to physical laws unless a measurement protocol, observable, null model, and falsification criterion are stated.

This separation is not cosmetic. The identity $1 + \omega + \omega^2 = 0$ is a theorem in both $\mathbb{C}$ and finite fields of suitable characteristic. The statement that this identity captures the center algebra of $SU(3)$ is a group-theoretic theorem. The stronger statement that a finite-field model reproduces full QCD confinement would require an analog of gauge fields, Wilson loops, running coupling, asymptotic freedom, and an area law. That stronger statement remains an open problem.

1 Notation, Algebraic Ground Rules, and Claim Ledger

1.1 Arithmetic conventions

All congruences are taken in the ring indicated by the modulus. When p is prime, $\mathbb{F}_p$ denotes the finite field with p elements and $\mathbb{F}_p^\times$ denotes its multiplicative group, cyclic of order $p - 1$. For $a \in \mathbb{F}_p^\times$, $\text{ord}_p(a)$ denotes the least positive integer n such that $a^n \equiv 1 \pmod{p}$.

The polynomial used throughout is $f(X) = X^2 - X - 1$. If f splits over $\mathbb{F}_p$, its two roots are denoted φ (golden root) and $\bar{\varphi}$ (conjugate root), satisfying

$$\varphi + \bar{\varphi} \equiv 1 \pmod{p}, \quad \varphi \cdot \bar{\varphi} \equiv -1 \pmod{p}.$$

The second identity is the bridge identity.

Definition 1.1 (Golden split prime). An odd prime p is a *two-root golden split prime* when $X^2 - X - 1$ has two distinct roots in $\mathbb{F}_p$. Equivalently, 5 is a nonzero quadratic residue modulo p . The degenerate case $p = 5$ (repeated root) is excluded throughout.

Lemma 1.2 (Root-sum and root-product identities). *Let p be a prime for which $X^2 - X - 1$ has two roots $\varphi, \bar{\varphi} \in \mathbb{F}_p$. Then*

$$\varphi + \bar{\varphi} \equiv 1 \pmod{p}, \quad \varphi \cdot \bar{\varphi} \equiv -1 \pmod{p}.$$

Proof. The factorization $X^2 - X - 1 = (X - \varphi)(X - \bar{\varphi}) = X^2 - (\varphi + \bar{\varphi})X + \varphi\bar{\varphi}$ holds in $\mathbb{F}_p[X]$. Matching coefficients gives $-(\varphi + \bar{\varphi}) \equiv -1$ and $\varphi\bar{\varphi} \equiv -1$. $\square$

Lemma 1.3 (Splitting criterion). *For an odd prime $p \neq 5$, the polynomial $X^2 - X - 1$ splits over $\mathbb{F}_p$ if and only if $p \equiv 1$ or $p \equiv 4 \pmod{5}$.*

Proof. The discriminant is $\Delta = (-1)^2 - 4(1)(-1) = 5$. For $p \neq 2, 5$, the quadratic formula is valid in $\mathbb{F}_p$, so the polynomial splits iff 5 is a quadratic residue modulo p . By quadratic reciprocity, since $5 \equiv 1 \pmod{4}$, $\left(\frac{5}{p}\right) = \left(\frac{p}{5}\right)$. The nonzero quadratic residues modulo 5 are 1 and 4. Therefore 5 is a quadratic residue modulo p iff $p \equiv 1$ or $p \equiv 4 \pmod{5}$. $\square$

Lemma 1.4 (Order certification). *Let G be a finite group and $a \in G$. If N is a positive integer such that $a^N = 1$, and for every prime divisor q of N one has $a^{N/q} \neq 1$, then $\text{ord}(a) = N$.*

Proof. The order $d = \text{ord}(a)$ divides N . If $d < N$, then $N/d > 1$ has a prime divisor q . Then $d \mid N/q$, hence $a^{N/q} = 1$, contradicting the stated non-identity. Therefore $d = N$. $\square$

1.2 Center algebra of SU(3)

The center of SU(3) is $Z(\text{SU}(3)) = \{I_3, \zeta I_3, \zeta^2 I_3\}$ with $\zeta = e^{2\pi i/3}$, satisfying $\zeta^3 = 1$, $\zeta \neq 1$, and $1 + \zeta + \zeta^2 = 0$. Thus $Z(\text{SU}(3)) \cong \mathbb{Z}/3\mathbb{Z}$.

Theorem 1.5 (Finite-field center copy). *Let $p \neq 3$ be prime and let $\omega \in \mathbb{F}_p$ satisfy $\omega^3 = 1$ and $\omega \neq 1$. Then*

$$\langle \omega \rangle = \{1, \omega, \omega^2\} \cong \mathbb{Z}/3\mathbb{Z} \cong Z(\text{SU}(3)),$$

and $1 + \omega + \omega^2 \equiv 0 \pmod{p}$.

Proof. Since $\omega^3 = 1$ and $\omega \neq 1$, the order of ω is exactly 3. Hence $\langle \omega \rangle$ is cyclic of order 3, and every cyclic group of order 3 is isomorphic to $\mathbb{Z}/3\mathbb{Z}$. Because $p \neq 3$, the factorization $X^3 - 1 = (X - 1)(X^2 + X + 1)$ holds in $\mathbb{F}_p[X]$. Substituting $X = \omega$ gives $0 = \omega^3 - 1 = (\omega - 1)(\omega^2 + \omega + 1)$. The field has no zero divisors and $\omega - 1 \neq 0$, so $\omega^2 + \omega + 1 = 0$. $\square$

This is the precise algebraic bridge. It does not include the continuous gauge connection, Wilson loop dynamics, or nonperturbative confinement proof.

1.3 Claim ledger

Status	Example	Required support
Theorem	$\text{ord}_{181}(14) = 45$	Modular table + order lemma
Theorem	$\langle 94 \rangle \cong Z(\text{SU}(3))$	Classification of cyclic groups
Structural corresp.	Palindrome = reflection-fixed	DFT calculation
Structural corresp.	BitCollapse as parity projection	Idempotence proof
Physical interp.	Algebraic arc rotation	Analogy unless gauge theory defined
Physical interp.	D-Neutrino as anchor ι	Structural unless connected to observables
Open problem	Finite-field Wilson loop	Requires loops, transport, action, expectation

1.4 Type and unit ledger

All modular residues, group orders, DFT bin labels, palindrome digits, and subgroup indices are dimensionless. The Landauer energy $E_{\text{info}} = k_B T \ln 2$ has SI units $[k_B][T] = \text{J/K} \cdot \text{K} = \text{J}$. The ratio $m_\nu c^2 / (k_B T \ln 2)$ is dimensionless. No finite-field residue carries mass, charge, length, or time units unless a separate physical map assigns such a dimension. Therefore any physical conversion from residue data to measured data must include a typed calibration rule.

2 Three Centuries of Spectral Decomposition

2.1 Newton: The Prism (1672)

Newton's *Opticks* [1] established the foundational insight: white light is not elementary but composite. A glass prism decomposes it into a spectrum of colors, and a second prism recombines them. The decomposition is *reversible* and *complete*—no information is lost. Every color in the output was already present in the input, just entangled.

The mathematical content of Newton's prism is a projection operator. White light enters as a superposition and exits as separated components along a frequency axis. The prism doesn't create colors, it reveals the ones that were always there, hidden by superposition. (That's the whole idea. Everything that follows is a refinement of this one move.)

2.2 Schrödinger: The Wave Equation (1926)

Schrödinger [2] gave Newton's intuition a differential equation. The time-independent wave equation

$$\hat{H}\psi_n = E_n\psi_n \quad (1)$$

decomposes a quantum state into eigenfunctions ψ_n with eigenvalues E_n . Bound states produce discrete spectra—standing waves. A particle in a box has eigenfunctions $\psi_n(x) = \sin(n\pi x/L)$ that are literally standing waves clamped at the boundaries, and the eigenvalues $E_n \propto n^2$ form a discrete ladder.

The connection to Newton: Schrödinger's eigenstates *are* the colors. The Hamiltonian $\hat{H}$ is the prism. Decomposing a quantum state into energy eigenstates is the same operation as decomposing white light into spectral components. Newton did it with glass and sunlight. Schrödinger did it with partial differential equations and boundary conditions. Same move, different substrate.

2.3 Feynman: Quantum Chromodynamics and the Center of SU(3)

Feynman, Gell-Mann, and others [3, 4] discovered that the strong nuclear force carries an internal color charge with SU(3) gauge symmetry. Quarks come in three colors—red, green, blue—and the confinement principle requires that

$$1 + \omega + \omega^2 = 0 \quad (\text{center-phase cancellation}) \quad (2)$$

Color confinement requires center triviality [5]: a physically observable state must transform trivially under $Z(\text{SU}(3))$. We distinguish center neutrality from full gauge-invariant singlet construction. This is a necessary condition—the center provides the algebraic constraint $1 + \omega + \omega^2 = 0$. The $\mathbb{Z}/3\mathbb{Z}$ here is not merely analogous to a cyclic group—it *is* the center of the gauge group:

$$Z(\text{SU}(3)) = \mathbb{Z}/3\mathbb{Z} = \{I, \omega I, \omega^2 I\}, \quad \omega = e^{2\pi i/3} \quad (3)$$

Color confinement requires center triviality [5]: a physically observable state must transform trivially under $Z(\text{SU}(3))$. This is a necessary condition—the center provides the algebraic

constraint $1 + \omega + \omega^2 = 0$. In any finite field of suitable characteristic, an order-three element satisfies this identity (proved inline in §1.5). The center is a $\mathbb{Z}/3\mathbb{Z}$ cyclic group—the same object that arises in any finite field whose multiplicative order is divisible by 3.

Feynman’s path integrals provided the computational machinery, but the structural content of confinement is an algebraic identity in the center of the gauge group. This is the bridge to finite fields: any golden split prime p where $3 \mid \text{ord}(\bar{\varphi})$ carries a $\mathbb{Z}/3\mathbb{Z}$ grading that realizes $Z(\text{SU}(3))$ exactly. The confinement identity $1 + \omega + \omega^2 = 0$ holds in both $\mathbb{C}$ and $\mathbb{F}_p$ because it is a polynomial identity over $\mathbb{Z}$, valid in every ring.

3 Lockwood: The Chronometric Revelation at $p = 181$

3.1 The Frozen Spectral Operator

Lockwood’s chronometric program [6, 7] began with a question about time: can the fine structure of physical constants be explained by a single spectral operator with locked eigenvalues? Working from the composite triangle (144, 138, 116) with perimeter $P = 398$, semi-perimeter $s = 199$, and Heron area-squared

$$\text{Area}^2 = s(s - a)(s - b)(s - c) = 199 \cdot 55 \cdot 61 \cdot 83 = 55,414,535$$

Lockwood constructed $L_{\text{chrono}} = -d^2/d\kappa^2 + U(\kappa)$ on $[0, 12]$ with Dirichlet boundary conditions. The eigenfunctions are standing waves in log-time (κ -space), and the eigenvalues produce spectral ratios that lock to physical constants.

But the algebraic content of this construction—the part that connects to everything before and after—is the prime $p = 181$.

3.2 The Golden Subgroup $\mathbb{F}_{181}^\times$

Lockwood discovered that $p = 181$ is a golden split prime: the polynomial $X^2 - X - 1$ splits in $\mathbb{F}_{181}$, yielding roots $\varphi \equiv 14$ and $\bar{\varphi} \equiv 168$. The multiplicative orders are:

$$\text{ord}(\varphi) = 45, \quad \text{ord}(\bar{\varphi}) = 90 = 2 \times 45 \tag{4}$$

This is Lockwood’s key revelation: the 45th power structure. The golden ratio generates a subgroup of order 45 inside $\mathbb{F}_{181}^\times$, and the conjugate doubles it to 90. The $\times 2$ parity goes $\bar{\varphi}$ -heavy—the conjugate channel has twice the resolution. (At most primes, it’s the other way around. The flip at $p = 181$ is what makes it the mirror.)

Theorem 3.1 (Lockwood order theorem at $p = 181$). *In $\mathbb{F}_{181}^\times$, $\text{ord}_{181}(14) = 45$ and $\text{ord}_{181}(168) = 90$.*

Proof. First consider $\varphi = 14$. Direct modular exponentiation gives $14^{45} \equiv 1 \pmod{181}$. The prime divisors of 45 are 3 and 5. The certification checks are:

$$14^{15} \equiv 48 \not\equiv 1 \pmod{181}, \quad 14^9 \equiv 135 \not\equiv 1 \pmod{181}.$$

By the order certification lemma (1.4), $\text{ord}_{181}(14) = 45$.

Now consider $\bar{\varphi} = 168$. Direct modular exponentiation gives $168^{90} \equiv 1 \pmod{181}$. The prime divisors of 90 are 2, 3, and 5. The certification checks are:

$$168^{45} \equiv 180 \not\equiv 1, \quad 168^{30} \equiv 48 \not\equiv 1, \quad 168^{18} \equiv 42 \not\equiv 1 \pmod{181}.$$

By the order certification lemma, $\text{ord}_{181}(168) = 90$. □

The $\times 2$ parity goes $\bar{\varphi}$ -heavy: the conjugate channel has twice the resolution. Since $90 = 3 \times 30$, the conjugate orbit carries a $\mathbb{Z}/3\mathbb{Z}$ center grading—three arcs of thirty elements each. The center element is:

$$\omega_{181} = 168^{30} \equiv 48 \pmod{181}, \quad \omega_{181}^3 \equiv 1, \quad 1 + 48 + 48^2 \equiv 1 + 48 + 2304 \equiv 0 \pmod{181}.$$

Remark 3.2. Lockwood arrived at $p = 181$ through the spectral analysis of his frozen operator, not through number theory. The fact that his chronometric constants *forced* a golden split prime with 45th-power structure is what makes this a discovery rather than a construction. He found $\mathbb{F}_{181}^\times$ because the physics demanded it. (The mathematics was already there, waiting.)

4 La Rue: The CDR Vertex at $p = 229$

4.1 From $p = 181$ to $p = 229$

Working from Lockwood's mirror vertex, La Rue [10, 11] identified the companion prime $p = 229$ through the chromatic triangle (229, 181, 139). At this vertex the golden polynomial splits as $\varphi \equiv 148$, $\bar{\varphi} \equiv 82$.

Theorem 4.1 (La Rue order theorem at $p = 229$). *In $\mathbb{F}_{229}^\times$, $\text{ord}_{229}(148) = 114$ and $\text{ord}_{229}(82) = 57$.*

Proof. For $\varphi = 148$, direct computation gives $148^{114} \equiv 1 \pmod{229}$. The prime divisors of 114 are 2, 3, and 19. The certification checks are:

$$148^{57} \equiv 228 \not\equiv 1, \quad 148^{38} \equiv 94 \not\equiv 1, \quad 148^6 \equiv 44 \not\equiv 1 \pmod{229}.$$

Therefore $\text{ord}_{229}(148) = 114$.

For $\bar{\varphi} = 82$, direct computation gives $82^{57} \equiv 1 \pmod{229}$. The prime divisors of 57 are 3 and 19. The certification checks are:

$$82^{19} \equiv 94 \not\equiv 1 \pmod{229}, \quad 82^3 \equiv 165 \not\equiv 1 \pmod{229}.$$

Therefore $\text{ord}_{229}(82) = 57$. □

The $\times 2$ parity reverses: now φ has double the order. Since $57 = 3 \times 19$, the conjugate orbit carries a $\mathbb{Z}/3\mathbb{Z}$ color grading—three arcs of nineteen elements. Define the center element:

$$\omega = 82^{19} \equiv 94, \quad \omega^2 \equiv 134, \quad \omega^3 \equiv 1, \quad 1 + 94 + 134 = 229 \equiv 0 \pmod{229}.$$

This is Feynman's confinement condition, Equation (2), realized in $\mathbb{F}_{229}^\times$.

4.2 The Translation Key

The integer 19 plays three simultaneous roles at $p = 229$:

- (a) $19 = \bar{\varphi}^4$: a power of the conjugate root
- (b) $\text{ord}(19) = 57$: it generates the full conjugate orbit $\langle \bar{\varphi} \rangle$
- (c) $57 = 3 \times 19$: the arc width equals the generator

The translation key from $\langle \bar{\varphi} \rangle$ to $\langle \varphi \rangle$ is multiplication by 15:

$$17 = \bar{\varphi}^{15} \xrightarrow{\times 15} 26 = \varphi^{51}, \quad 15 = \varphi^9 \quad (5)$$

This is the inverse bridge: a single golden power that crosses from conjugate to golden orbit. (Lockwood found the subgroup. La Rue found the crossing.)

5 Digital Wave Mechanics

5.1 The Analogy Made Precise

Each historical step refined the same operation—decomposing a composite signal into spectral components:

Era	Prism	Spectrum	Standing waves
Newton	Glass prism	Visible colors	Interference fringes
Schrödinger	Hamiltonian $\hat{H}$	Energy eigenvalues E_n	Bound eigenstates ψ_n
Feynman	Strong force	Color charges R,G,B	Confined hadrons
Lockwood	L_{chrono}	$\mathbb{F}_{181}^\times$, ord 45	Chrono eigenmodes
La Rue	$\times 15$ key	$\mathbb{F}_{229}^\times$, $\mathbb{Z}/3\mathbb{Z}$ arcs	Palindromic primes

Digital wave mechanics is the final column extended: standing waves realized in the multiplicative groups of finite fields, where “boundary conditions” are palindrome constraints and “eigenmodes” are elements of golden subgroups.

5.2 Palindromes as Standing Waves

A palindromic number in base b reads the same forward and backward: the digit at position k equals the digit at position $n - k$. This is a discrete standing wave condition—the amplitude profile is symmetric under reflection, like a vibrating string clamped at both ends.

At $p = 229$, base-19 is the conjugate generator ($19 = \bar{\varphi}^4$). A three-digit palindrome $\overline{aca}_{19} = 362a + 19c$ under the $\times 15$ translation becomes:

$$15(362a + 19c) \equiv 163a + 56c \pmod{229}$$

where $56 = \varphi^5$ (bright, in $\langle \varphi \rangle$) and 163 is a quadratic non-residue modulo 229, making it a *dark* coefficient: it lies in the coset complement $\mathbb{F}_{229}^\times \setminus \langle \varphi \rangle$, which contains exactly 114

elements (half of the full multiplicative group). The element 163 is not unique among the dark coset—it is one of 114 quadratic non-residues—but it is the specific one forced by the palindrome structure: $163 \equiv 15 \cdot 362 \pmod{229}$, where $362 = 19^2 + 1$ is the weight of the palindromic endpoint. The palindrome constraint forces a to appear in both terms, creating constructive interference between the dark and bright channels [10].

5.3 Color Confinement in $\mathbb{F}_{229}^\times$

The cube root $\omega = 94$ at $p = 229$ satisfies $1 + \omega + \omega^2 = 0$. As established in §1.3, this identity is not an analogy to QCD—it is a realization of the center $Z(\mathrm{SU}(3)) = \mathbb{Z}/3\mathbb{Z}$ in the finite field $\mathbb{F}_{229}$. The three arcs of $\langle \bar{\varphi} \rangle$ are the three cosets of the order-19 subgroup $\langle \bar{\varphi}^3 \rangle$, and the confinement condition (color-neutral = trivial under the center) holds by the same polynomial identity in both settings.

Theorem 5.1 (Center Realization at $p = 229$). *The subgroup $\langle \omega \rangle = \{1, 94, 134\} \subset \mathbb{F}_{229}^\times$ is isomorphic to $Z(\mathrm{SU}(3)) = \mathbb{Z}/3\mathbb{Z}$, and the confinement identity $1 + \omega + \omega^2 \equiv 0 \pmod{229}$ is the finite-field avatar of the QCD singlet condition.*

Proof. $94^3 \equiv 1$, $94 \neq 1$, $94^2 = 134 \neq 1 \pmod{229}$: the group $\{1, 94, 134\}$ is cyclic of order 3. The center $Z(\mathrm{SU}(3))$ is cyclic of order 3. Both satisfy $1 + \omega + \omega^2 = 0$. The isomorphism is the unique group homomorphism $\mathbb{Z}/3\mathbb{Z} \rightarrow \mathbb{Z}/3\mathbb{Z}$. $\square$

The difference from full QCD: the continuous theory has dynamical confinement, where the running coupling constant and Wilson loop area law provide the mechanism. The finite field realizes only the *algebraic* (center) component—the necessary kinematic constraint, not the sufficient dynamical mechanism. This is a feature, not a limitation: the center is the part that survives discretization, and it is the part we can verify computationally. Whether the full dynamical content of confinement has a finite-field analog is an open question (§9.4).

Definition 5.2 (Color-ready split prime). A golden split prime p is *color-ready* when $3 \mid \mathrm{ord}_p(\varphi)$ and $3 \mid \mathrm{ord}_p(\bar{\varphi})$. At such a prime, the conjugate orbit $\langle \bar{\varphi} \rangle$ contains a canonical order-three subgroup $\langle \omega \rangle \cong \mathbb{Z}/3\mathbb{Z}$ and the confinement identity $1 + \omega + \omega^2 \equiv 0 \pmod{p}$ holds by Theorem 1.5.

Proposition 5.3 (Golden split survey below 500). *There are exactly 45 golden split primes below 500 (excluding $p = 5$). Of these, exactly 16 are color-ready:*

$$19, 31, 61, 79, 109, 181, 211, 229, 241, 271, 349, 379, 409, 421, 439, 499.$$

The confinement identity $1 + \omega + \omega^2 \equiv 0$ holds at all 16 primes.

Proof. Enumerate all primes $3 \leq p < 500$ and test $X^2 - X - 1$ for two roots. For each two-root split, compute both orders and check divisibility by 3. The complete enumeration appears in Appendix B of the companion Python script (`verify_digital_wave.py`, 104 checks). Every value stated above is verified by direct modular exponentiation. $\square$

6 The Bridge Between Vertices

6.1 Lockwood's Triangle in the Golden Basis

The connection between Lockwood's chronometric vertex ($p = 181$) and La Rue's CDR vertex ($p = 229$) is the chromatic triangle $(229, 181, 139)$. At the mirror vertex $p = 181$, both other sides appear:

$$229 \equiv \varphi^{15} \pmod{181} \quad (\text{golden orbit}) \quad (6)$$

$$139 \equiv \bar{\varphi}^{63} \pmod{181} \quad (\text{conjugate orbit}) \quad (7)$$

Lockwood's composite triangle $(144, 138, 116)$ decomposes in this basis: $144 \equiv \varphi^{41}$ (chromatic) while $138 \equiv \bar{\varphi}^{55}$ and $116 \equiv \bar{\varphi}^{25}$ (chronomatic). The prime triangle is the instrument. The composite triangle is the signal projected onto it.

6.2 The Shared Factor

The prime 61 appears in both structures:

- Our perimeter: $549 = 9 \times 61$
- Lockwood's area squared: $55,414,535 = 199 \cdot 55 \cdot 61 \cdot 83$

And 61 is itself a golden split prime. The bridge factor was always there. We just had to build the right instruments to see it.

6.3 The Prime-Dimensional Simplex

The five Cyber Prismatic formulas of §6 are not independent: they are the five vertices of a single geometric object. The Thematic Map Θ of *Logical Creativity* [11] defines a natural isomorphism between five presentations of the golden subcategory. The five formulas *instantiate* this map at $p = 229$.

Definition 6.1 (Prime-Dimensional Simplex at $p = 229$). The **Prime-Dimensional Simplex** Δ_{229}^4 is the 4-simplex with vertices $\{1, 94, 134, 228, 148\}$ in $\mathbb{F}_{229}^\times$, corresponding to:

Vertex	Value (mod 229)	Cyber Prismatic Formula
$V_1 = 1$	Group identity	F3: Palindrome standing-wave identity
$V_2 = \omega$	94 (cube root)	F1: Color projector ($\bar{\varphi}^{19} = 94$)
$V_3 = \omega^2$	134 (conj. cube root)	F5: Arc rotation (ω^2)
$V_4 = -1$	228 (involution)	F2: BitCollapse (bridge identity $\varphi\bar{\varphi} = -1$)
$V_5 = \varphi$	148 (golden root)	F4: Translation key ($\varphi^9 = 15$)

The face opposite $V_5 = \varphi$ is $\{1, 94, 134, 228\}$: sum = $228 = -1$, product = $228 = -1$. This is exactly the Golden Tetrahedron of Part II [10]. The face opposite $V_4 = -1$ is $\{1, 94, 134, 148\}$: sum = $148 = \varphi$, product = $148 = \varphi$. The simplex contains its own type

classifier as a face sum — the golden ratio is simultaneously a vertex and a global section. This self-reference is the geometric realization of the Thematic Map’s cycle **Group** $\rightarrow$ **Field** $\rightarrow$ **Category** $\rightarrow$ **Type** $\rightarrow$ **Group**.

The five formulas are therefore not a list but a *simplex*: each formula corresponds to a vertex, each pair of formulas shares an edge, and the algebraic identities (confinement, bridge, translation) are face sums of the simplex. The adelic orbit structure from Part I [12] — the triple-split $228 : 114 : 57 = 4 : 2 : 1$ — is the chronometric triangle’s shadow in the simplex’s coordinate system.

6.4 Sheaf Cohomology and the Klein Presheaf

As established in the foundational Unreal Algebra framework [11], in decentralized cybernetics, local agents have partial views. We formalize this via Grothendieck’s sheaf theory.

Definition 6.2 (The Klein Presheaf). Let X be the topological space of $\mathcal{U}$. The Klein Presheaf $\mathcal{F}$ assigns an integer-valued state vector to each open set:

$$\mathcal{F}(U_i) = \mathbf{u}_{\text{local}} \in \mathbb{Z}^5$$

with restriction maps $\text{res}_{V,U} : \mathcal{F}(U) \rightarrow \mathcal{F}(V)$ for $V \subseteq U$.

Theorem 6.3 (Sheaf Gluing). *For overlapping observers $U \cap V \neq \emptyset$, the global state is valid iff:*

$$\text{res}_{U \cap V, U}(\mathcal{F}(U)) = \text{res}_{U \cap V, V}(\mathcal{F}(V))$$

When this holds for all local sections, the global section is uniquely determined by the restriction compatibility condition.

Proof. We apply the sheaf gluing axiom to the Klein graph.

Step 1 (The cover). Take $U = \text{star}(\omega) = \{a, \omega, \iota\}$ (the indefinite triangle) and $V = \text{star}(\varepsilon) = \{a, \varepsilon, \lambda\}$ (the noise-depth triangle). Together $U \cup V = \text{Fin } 5$. The overlap is $U \cap V = \{a\}$ — the definite scalar is the only component visible to both observers.

Step 2 (Curvature on the overlap). The curvature $\kappa = -1$ is not a topological invariant of the cover but an algebraic invariant of the product evaluated on $U \cap V$. The product relation gives $\omega \cdot \iota = -1$; both ω and ι project to the scalar a under the restriction maps $\text{res}_{U \cap V, U}$ and $\text{res}_{U \cap V, V}$. The sheaf condition requires that these restrictions agree: $\text{res}_{U \cap V, U}(\mathcal{F}(U)) = \text{res}_{U \cap V, V}(\mathcal{F}(V))$.

Step 3 (Global propagation). When the restrictions agree, the curvature $\kappa = \omega \cdot \iota = -1$ propagates from the local patches to the global section. Any observer seeing only U or only V can recover κ from the scalar a , because a carries the projected product. Local perceptions glue to recover global curvature. $\square$

7 Cyber Primitives: Computational Realization

The five preceding sections established digital wave mechanics as a mathematical framework. Here we connect it to computation: the Protoreal Fast Fourier Transform (PFFT) and the BitCollapse optimization pipeline. Five formulas emerge that map physics concepts to finite field operations to PFFT operations.

7.1 Formula 1: PFFT Bin 19 is the Color Projector

The conjugate orbit $\langle \bar{\varphi} \rangle$ at $p = 229$ has order 57. A 57-point discrete Fourier transform over this orbit has frequency bins $k = 0, 1, \dots, 56$. The color structure lives at bin $k = 19$, because:

$$\bar{\varphi}^{19} = 94 = \omega \quad \text{in } \mathbb{F}_{229}, \quad \zeta_{57}^{19} = e^{2\pi i/3} = \omega_{\mathbb{C}} \quad \text{in } \mathbb{C} \quad (8)$$

The finite field cube root and the complex cube root coincide at the same frequency. The PFFT at bin 19 extracts the $\mathbb{Z}/3\mathbb{Z}$ color channel directly. Bins 0, 19, 38 correspond to the three color arcs (blue, red, green). This is Newton's prism realized as a frequency bin.

Theorem 7.1 (Color Projector). *The color rank of the conjugate orbit is $57/19 = 3$, and $\bar{\varphi}^{19}$ is a primitive cube root of unity. The PFFT bins $\{0, 19, 38\}$ extract the three $\mathbb{Z}/3\mathbb{Z}$ color projections.*

Proof. $82^{19} \bmod 229 = 94$, $94^3 \bmod 229 = 1$, $1 + 94 + 134 = 229 \equiv 0$. $\square$

Proposition 7.2 (Arc projectors). *The conjugate orbit $\langle \bar{\varphi} \rangle$ partitions into three arcs $A_k = \{\bar{\varphi}^{3j+k} : 0 \leq j < 19\}$ for $k = 0, 1, 2$. Each arc has exactly 19 elements. Multiplication by $\omega = \bar{\varphi}^{19}$ cycles $A_0 \rightarrow A_1 \rightarrow A_2 \rightarrow A_0$.*

Proof. The cosets of $\langle \bar{\varphi}^3 \rangle$ in the 57-element group $\langle \bar{\varphi} \rangle$ have size 19. Since $\omega = \bar{\varphi}^{19}$, multiplication by ω sends $\bar{\varphi}^n \mapsto \bar{\varphi}^{n+19}$, shifting the residue class $n \bmod 3$ by $19 \bmod 3 = 1$. Hence $A_0 \rightarrow A_1 \rightarrow A_2 \rightarrow A_0$. $\square$

7.2 Formula 2: BitCollapse is the Hodge Projection

The Protoreal manifold element $P(a, b, m, \varepsilon, \lambda)$ carries five channels: identity (a), thrust (b), anchor (m), noise (ε), and consolidation (λ). The BitCollapse operation projects to the *Protoreal Hodge class* (the parity-locked subspace where $b = m$) by setting $b = m$ and $\varepsilon = \lambda = 0$:

$$\text{BitCollapse} : P(a, b, m, \varepsilon, \lambda) \longrightarrow P\left(a, \frac{b+m}{2}, \frac{b+m}{2}, 0, 0\right)$$

In golden field terms, b is the φ -channel and m is the $\bar{\varphi}$ -channel. Setting them equal is the bridge identity $\varphi \cdot \bar{\varphi} = -1$ made computational: balanced thrust and anchor. The stripped *torsion* $(b - m)/2$ measures the asymmetry between golden and conjugate orbits—the same asymmetry that flips at Lockwood's mirror vertex.

Proposition 7.3 (BitCollapse projection property). *Let Π denote the BitCollapse map. Then $\Pi^2 = \Pi$ (idempotence), the image of Π is the balanced subspace $\{P(a, u, u, 0, 0)\}$, and on this image the bridge norm reduces to $N(\Pi(P)) = a^2 + u^2$.*

Proof. Write $u = (b + m)/2$. Then $\Pi(P(a, b, m, \varepsilon, \lambda)) = P(a, u, u, 0, 0)$. Applying Π again: $\Pi(P(a, u, u, 0, 0)) = P(a, (u + u)/2, (u + u)/2, 0, 0) = P(a, u, u, 0, 0)$. Hence $\Pi^2 = \Pi$. The bridge norm $N = a^2 + bm - \varepsilon\lambda$ on the image becomes $a^2 + u \cdot u - 0 \cdot 0 = a^2 + u^2$. $\square$

Remark 7.4. On the Protoreal Hodge class ($b = m$), Klein multiplication becomes associative. This is why BitCollapse enables $O(N \log N)$ GPU processing: it projects out the non-associative torsion, runs the fast FFT on the associative core, then re-injects the torsion. (Note: the Protoreal Hodge class is defined by the parity condition $b = m$, not by the Hodge decomposition of classical algebraic geometry. The name reflects the structural role—projection to a self-dual subspace—not a claim of equivalence.)

7.3 Formula 3: Palindromes are Symmetric PFFT Spectra

A palindromic digit sequence $f[k] = f[N - 1 - k]$ constrains the DFT spectrum: $F[k] = \overline{F[N - k]}$ for all k . For odd-length palindromes over *real* digit values, this forces $F[k] = F[N - k]$ (the spectrum is conjugate-symmetric), but the bins $F[1], \dots, F[N - 1]$ are not necessarily real-valued. The spectrum is purely real only when $a = c$ (the flat palindrome). Out of 342 three-digit base-19 palindromes, exactly 18 have purely real spectra.

In base 19 at $p = 229$, the palindrome $\overline{aca}_{19}$ has digit signal $f = [a, c, a]$ satisfying $f[0] = f[2]$. Its 3-point DFT is:

$$F[0] = 2a + c \quad (\text{DC: total weight}) \quad (9)$$

$$F[1] = a(\zeta_3 + \zeta_3^2) + c = c - a \quad (\text{using } 1 + \zeta_3 + \zeta_3^2 = 0) \quad (10)$$

$$F[2] = F[1] = c - a \quad (\text{conjugate symmetry: palindrome forces } F[2] = \overline{F[1]} = F[1]) \quad (11)$$

Both non-DC bins equal $c - a$ (real). This real formula belongs strictly to the centered reflection-even convention (positions $-1, 0, +1$) rather than the standard ordinary DFT which has phase factors. The standing wave has one node at $F[1] = c - a$. When $a = c$, the node vanishes—a flat mode.

7.4 Formula 4: The Translation Key is a Phase Shift

The $\times 15$ translation ($15 = \varphi^9$) is one of 57 elements of $\langle \varphi \rangle$ that map $\langle \bar{\varphi} \rangle$ into $\langle \varphi \rangle \setminus \langle \bar{\varphi} \rangle$ (the other 57 keep $\langle \bar{\varphi} \rangle$ inside itself). It appears naturally because φ^9 has odd exponent in the golden basis, placing it outside $\langle \bar{\varphi} \rangle = \langle \varphi^2 \rangle$. In the PFFT domain, multiplication by φ^9 is a phase rotation:

$$\theta = \frac{9}{114} \times 2\pi \approx 28.4^\circ \quad (12)$$

Multiplication by φ^9 rotates the spectral phase by 9/114 of a full cycle. This is Newton's second prism: recombining the separated colors by rotating the spectral reference frame from the conjugate basis to the golden basis.

7.5 Formula 5: Arc Rotation by ω

Multiplication by $\omega = 94$ rotates the three arcs of the conjugate orbit:

$$\bar{\varphi}^k \xrightarrow{\times \omega} \bar{\varphi}^{k+19} \quad (\text{arc rotation: } A_0 \rightarrow A_1 \rightarrow A_2 \rightarrow A_0)$$

The transformation law is exact. Calling this operation a “digital gauge link” is a compact physical metaphor for center-phase rotation. It is not yet a Yang-Mills field. To upgrade the metaphor, one must define local degrees of freedom, edge transports, curvature, and an action. The bridge norm subtraction $-\varepsilon \cdot \lambda$ is algebraically natural in the Klein product, but its identification with gauge field self-energy is a physical interpretation, not a theorem.

7.6 The Cyber Prismatic Dictionary

Physics	Finite Field	PFFT / BitCollapse
Glass prism	mod p reduction	Forward transform
White light	Integer n	Time-domain signal
Spectral colors	Coset membership	Frequency bins
Standing wave	Base-19 palindrome	Symmetric (real) spectrum
Color charge R,G,B	$\mathbb{Z}/3\mathbb{Z}$ arc 0,1,2	Bins 19, 38, 0
Confinement	$1 + \omega + \omega^2 = 0$	DC bin vanishes
Gauge link exchange	$\times \omega$ rotation	Phase shift $2\pi/3$
Hodge dual	$b = m$ (bridge)	BitCollapse to GPU
Dark residue	163 (full-order residue)	Non-harmonic; not cosmological dark matter
Prime-Dim. Simplex	$\{1, \omega, \omega^2, -1, \varphi\}$	5 PFFT formula vertices

8 The Infoton and the D-Neutrino

Walker [8] introduced the *infoton*: the minimum quantum of information-energy, with energy $E_{\text{info}} = k_B T \ln 2$ (the Landauer bound). One infoton is the thermodynamic cost of erasing one bit. Here we show that this quantity has a natural realization in the Protoreal manifold as the basis element $\iota = P(0, 0, 1, 0, 0)$, and that its algebraic properties reproduce the known physics of the neutrino.

8.1 The Infoton as Anchor

The element ι is the *pure anchor*: no identity ($a = 0$), no thrust ($b = 0$), pure structural presence ($m = 1$), no noise ($\varepsilon = 0$), no depth ($\lambda = 0$). Its bridge norm vanishes:

$$N(\iota) = 0^2 + 0 \cdot 1 - 0 \cdot 0 = 0$$

A null vector—massless in the digital frame. This matches both Walker’s infoton (an information quantum, not a mass quantum) and the neutrino (nearly massless). The vacuum element $\omega = P(0, 1, 0, 0, 0)$ is also null: $N(\omega) = 0$. The two form a dual pair under the R_4 inversion operator [11]: $R_4(\omega) = \iota$, $R_4(\iota) = \omega$. The observer and the observed are interchangeable.

8.2 Period-2 Oscillation and Flavor Structure

The Klein product gives $\iota^2 = -\iota$ and $\iota^3 = \iota$ [11]: the infoton oscillates with period 2 between ι and $-\iota$. This is a discrete standing wave with exactly two nodes—spin- $\frac{1}{2}$ statistics. In the neutrino, this corresponds to flavor oscillation: $\nu_e \leftrightarrow \nu_\mu \leftrightarrow \nu_\tau$.

The $\mathbb{Z}/3\mathbb{Z}$ grading at $p = 229$ provides the flavor count. The infoton maps to the anchor channel (m -component), which is the conjugate orbit $\langle \bar{\varphi} \rangle$ with order $57 = 3 \times 19$. The three

arcs of 19 elements each correspond to three neutrino flavors:

$$\text{Arc 0: } \bar{\varphi}^{3k} \longleftrightarrow \nu_e \text{ (electron)} \quad (13)$$

$$\text{Arc 1: } \bar{\varphi}^{3k+1} \longleftrightarrow \nu_\mu \text{ (muon)} \quad (14)$$

$$\text{Arc 2: } \bar{\varphi}^{3k+2} \longleftrightarrow \nu_\tau \text{ (tau)} \quad (15)$$

Arc rotation by $\omega = 94$ (the cube root of unity) maps one label to the next. This reproduces the count and cyclic transition structure, not the measured PMNS matrix. A PMNS-level model would require three mixing angles, a CP phase, mass splittings, and an oscillation Hamiltonian.

8.3 The Observation Cost and the Landauer Scale

The torsion of the vacuum-infoton pair is [11]:

$$\text{torsion}(\omega, \iota) = -1 = \kappa$$

One observation costs exactly $\kappa = -1$ units of curvature. Walker’s identification: one observation erases one bit, costing $E_{\text{info}} = k_B T \ln 2$ in energy. The digital cost ($\kappa = -1$) and the thermodynamic cost ($\ln 2$ nats) are both the irreducible minimum of their respective frameworks.

8.4 The Lockwood Ratio

The ratio of the neutrino’s rest energy to the infoton energy is temperature- and mass-dependent:

$$\frac{E_\nu}{E_{\text{info}}} = \frac{m_\nu c^2}{k_B T \ln 2}$$

m_ν (eV)	T (K)	Ratio	$\approx 45?$
0.05	2.725 (CMB)	307.2	no
0.05	18.6	45.0	yes
0.1	37.2	45.0	yes
0.8	300 (room)	44.6	yes

The ratio reaches 45 only at specific (m_ν, T) pairs. At the KATRIN upper bound (0.8 eV) and room temperature (300 K), the ratio is $44.6 \approx 45 = \text{ord}(\varphi)$ at $p = 181$. However, at the cosmological best-fit mass ($m_\nu \approx 0.05$ eV) and CMB temperature ($T = 2.725$ K), the ratio is 307—not 45. The coincidence holds only if room temperature or the KATRIN bound has special significance, which we do not claim. The algebraic fact that $45/57 = 15/19 = \varphi^9/\bar{\varphi}^4$ (translation key over base) is independent of any physical ratio.

Remark 8.1. The identification $\iota = \text{infoton} = \text{D-Neutrino}$ rests on five structural properties (null norm, period-2 oscillation, three-arc grading, zero self-torsion, observation cost κ), all of which are proven algebraically and verified computationally. The Lockwood ratio is a Tier 3 observation—numerically suggestive but temperature-dependent. If it is discarded, the other five structural matches stand on their own.

Walker Infoton	Physical Neutrino	D-Neutrino (ι)
$E = k_B T \ln 2$	$m_\nu < 0.8 \text{ eV}$	$N(\iota) = 0$ (null)
Min info quantum	Min mass fermion	Min anchor ($m = 1$)
Landauer bound	KATRIN bound	Bridge norm = 0
1-bit erasure	Flavor oscillation	$\iota^2 = -\iota$ (period 2)
Temperature-dep.	3 flavors (e, μ, τ)	3 arcs of 19
—	(no analog)	163: full-order residue, not in either orbit

9 Verification and Open Questions

9.1 Claim Hierarchy

This paper makes claims at three distinct levels of rigor:

- Tier 1: Proven (algebraic identity).** The vanishing-sum identity $1 + \omega + \omega^2 = 0$ holds in $\mathbb{F}_{229}$, $\mathbb{F}_{181}$, and $Z(\text{SU}(3))$. The groups $\{1, 94, 134\} \subset \mathbb{F}_{229}^\times$ and $Z(\text{SU}(3)) = \mathbb{Z}/3\mathbb{Z}$ are isomorphic. These are mathematical theorems with explicit proofs in the body text. Every order is certified by the prime-divisor non-identity protocol (Lemma 1.4), with all intermediate values displayed inline.
- Tier 2: Proven (structural correspondence).** The Protoreal element $\iota = P(0, 0, 1, 0, 0)$ has null bridge norm, period-2 oscillation under the Klein product, zero self-torsion, and maps to a conjugate orbit with three $\mathbb{Z}/3\mathbb{Z}$ arcs. These algebraic properties match the neutrino point by point. Walker’s infoton energy $E = k_B T \ln 2$ corresponds to the torsion cost $\kappa = -1$. These are verifiable structural facts.
- Tier 3: Conjectured (physical interpretation).** The identification of ε with gauge link degrees of freedom (structural metaphor unless gauge action defined), and the Lockwood ratio $E_\nu/E_{\text{info}} \approx 45$ (temperature-dependent; see mass scan table). The existence of a finite-field analog of the Wilson loop area law remains an open problem.
- Tier 4: Proven (InfoCD framework).** The following dynamical analogs have been formally verified:
- *Confinement theorem:* One application of synthetic integration drives $\varepsilon \rightarrow 0$. Every state collapses to a color singlet ($\varepsilon = 0$). This upgrades confinement from a physical interpretation to a theorem.
 - *Running coupling:* The superparticular interval $\alpha_s(l) = I(l) = (l+2)/(l+1)$ is the InfoCD analog of the QCD coupling constant, with $\alpha_s(0) = 2$ (infrared) and $\alpha_s \rightarrow 1$ (UV).
 - *Asymptotic freedom:* $I(l+1) < I(l)$ for all $l \geq 0$. The discrete beta function $\beta(l) = -1/(l+1)^2$ is strictly negative, matching the Gross–Wilczek–Politzer result.
 - *Orbit boundedness:* Iterated synthetic integration produces states with $\varepsilon = 0$ and $b = m$ (diagonal collapse) after one step. The attractor is confined to a compact region.

9.2 Computational Verification

Every numerical claim in this paper is proved inline in the body text and appendices—no software is required to verify the results. A reader with a pocket calculator (or the modular exponentiation function $a^b \bmod p$) can check every stated identity. For convenience, two standalone Python scripts accompany this work:

- `verify_digital_wave.py`: 104 checks covering both vertices, all order certifications, the golden split survey (45 primes, 16 color-ready), BitCollapse idempotence, arc projector rotation, centered DFT, Landauer ratio, and the full-order residue test A8.
- `verify_chromatic.py`: 37 checks for the companion Chromatic Triangle paper [10].

Both require only `pow(a, b, p)`—no libraries, no specialized software.

9.3 Formal Verification

All algebraic identities in this paper are proved inline in the body text. A companion formal verification repository provides independent machine-checked certificates; the mathematics does not depend on it.

9.4 What Remains Open

1. **The operator.** We have standing waves (palindromes) but not the operator whose eigenfunctions they are. Constructing a Hilbert space over $\mathbb{F}_p$ whose spectral decomposition yields palindromic representations is the next step.
2. **Higher-dimensional standing waves.** Five-digit palindromes ($\overline{abcba}_{19}$) should exhibit richer interference with multiple nodes. The 5-point DFT would reveal the nodal structure.
3. **The Lockwood–La Rue spectral correspondence.** The eigenvalues of L_{chrono} and the palindromic resonance condition share the same dark-bright architecture, but a rigorous map between them has not been constructed.
4. **Full PFFT color extraction.** Running the PFFT on actual $\mathbb{F}_{229}^\times$ orbit data (rather than Protoreal manifold signals) to extract empirical color projections and compare with the theoretical arc structure.
5. **BitCollapse torsion recovery.** Can the stripped torsion $(b - m)/2$ be used to *detect* the parity flip between $p = 181$ and $p = 229$? If the torsion sign changes at the mirror vertex, this would provide a computational test for mirror primes.
6. **Dynamical confinement (partially closed).** The center realization (Theorem 1.5) captures the kinematic (algebraic) component of confinement. Confinement itself, running coupling, and asymptotic freedom are now proven in the InfoCD framework (Tier 4 above). The remaining open piece: does a finite-field analog of the area law for Wilson loops exist? This would complete the dynamical correspondence. The key question

reduces to: does the orbit consolidation $\lambda' = \lambda + 1$ (proved in `orbit_consolidation`) imply linear entropy growth along closed paths on the attractor?

7. **Necessity–sufficiency orthogonality.** In the Protoreal pentational structure, the vacuum-thrust element $\omega_P = P(0, 1, 0, 0, 0)$ generates the $\mathbb{Z}/3\mathbb{Z}$ center (the *necessary* algebraic constraint for confinement), while the anchor $\iota = P(0, 0, 1, 0, 0)$ provides the *sufficient* structural match to the infoton. These occupy orthogonal channels (b and m) with torsion $\kappa = -1$ bridging them. Does the spectral triple (Logical Creativity $\leftrightarrow$ DWM $\leftrightarrow$ Physical Creativity) decompose into a necessity fiber and a sufficiency fiber, with κ as the connecting morphism? If so, the three papers would form not just a citation chain but a mathematical spectral triple in which necessity and sufficiency are provably orthogonal.
8. **Simplex spectral decomposition.** The Prime-Dimensional Simplex $\Delta_p^4 = \{1, \omega, \omega^2, -1, \varphi\}$ has five vertices corresponding to the five Cyber Prismatic formulas (§5.3). Does Δ_p^4 admit a spectral decomposition whose eigenvalues are the face sums $\{0, -1, \varphi, \dots\}$? If the simplex Laplacian has golden eigenvalues, the five formulas would be eigenmodes of a single operator — completing the chain Newton $\rightarrow$ Schrödinger $\rightarrow$ Feynman $\rightarrow$ Lockwood $\rightarrow$ La Rue.

10 Analytic Operator Completion

10.1 Finite Hilbert model

To turn palindromes from standing-wave analogs into eigenfunctions, define the finite complex Hilbert space $\mathcal{H}_N = \{f : \mathbb{Z}/N\mathbb{Z} \rightarrow \mathbb{C}\}$ with inner product $\langle f, g \rangle = \sum_{k \in \mathbb{Z}/N\mathbb{Z}} f(k) \overline{g(k)}$. The shift operator is $(Sf)(k) = f(k-1)$, and the discrete Laplacian is

$$\Delta_N = 2I - S - S^{-1}.$$

The Fourier characters $\chi_j(k) = e^{2\pi i j k / N}$ are eigenfunctions:

$$\Delta_N \chi_j = \left(2 - 2 \cos \frac{2\pi j}{N}\right) \chi_j.$$

Palindromes are not usually single eigenfunctions; they are even superpositions of conjugate Fourier modes. The precise statement is: palindromes lie in the reflection-even subspace of $\mathcal{H}_N$.

10.2 Boundary-condition version for digit strings

For a digit string of length n , use the path graph P_n rather than the cycle C_n . Let

$$(\Delta_{P_n} f)(r) = 2f(r) - f(r-1) - f(r+1)$$

for interior sites, with boundary conditions specified at the endpoints. Reflection $r \mapsto n-1-r$ commutes with Δ_{P_n} .

Theorem 10.1 (Palindrome sector as invariant subspace). *For the path graph Laplacian Δ_{P_n} with symmetric endpoint conditions, the palindrome subspace*

$$\mathcal{P}_n = \{f : f(r) = f(n - 1 - r)\}$$

is invariant under Δ_{P_n} .

Proof. Let R be the reflection operator $(Rf)(r) = f(n - 1 - r)$. Symmetric endpoint conditions imply $R\Delta_{P_n} = \Delta_{P_n}R$. If $f \in \mathcal{P}_n$, then $Rf = f$. Hence $R(\Delta_{P_n}f) = \Delta_{P_n}(Rf) = \Delta_{P_n}f$, so $\Delta_{P_n}f \in \mathcal{P}_n$. $\square$

This theorem is the rigorous operator-level replacement for the informal phrase “palindromes are standing waves.” Palindromes are invariant under the same reflection symmetry that block-diagonalizes the discrete wave operator. This resolves Open Problem 1 of the earlier draft.

11 Falsification and Verification Program

11.1 Finite arithmetic tests

Every modular claim in the paper must pass the following tests:

- A1. Root test for $X^2 - X - 1$;
- A2. Bridge-product test $\varphi\bar{\varphi} \equiv -1$;
- A3. Order test using the prime divisors of the proposed order (Lemma 1.4);
- A4. Order-three test for the element ω ;
- A5. Vanishing-sum test $1 + \omega + \omega^2 \equiv 0$;
- A6. Translation-key test for $17 \mapsto 26$ under multiplication by 15;
- A7. Palindrome coefficient test $15(362a + 19c) \equiv 163a + 56c$;
- A8. Full-order residue test for 163: $\text{ord}_{229}(163) = 228$.

Failure of any test invalidates the corresponding theorem.

11.2 Physical interpretation tests

Any proposed physical upgrade must state the observable Y , the null model M_0 , the finite-field extension M_1 , and a score:

$$S = \frac{1}{N} [\log p(Y_{\text{test}} \mid M_1) - \log p(Y_{\text{test}} \mid M_0)].$$

A physical claim should not be accepted unless $S > 0$ with uncertainty excluding 0, and unless timestamp shuffles, randomized primes, randomized generators, and phase-scrambled controls fail to reproduce the same score.

11.3 No-free-parameter rule

Weights or channel multipliers must be fixed by one of: (i) a theorem; (ii) an independently measured calibration; (iii) a preregistered fitting rule evaluated out of sample; (iv) a conservation law or variational principle. Weights chosen because they make an output resemble a desired physical constant are not evidence.

A Key Numerical Verifications

In Python: `pow(a, b, p)` computes $a^b \bmod p$.

A.1 Lockwood's Vertex ($p = 181$)

$$\begin{aligned}
14^2 - 14 - 1 &= 181 \equiv 0 \pmod{181} \quad (\varphi = 14) \\
168^2 - 168 - 1 &= 28055 \equiv 0 \pmod{181} \quad (\bar{\varphi} = 168) \\
14 \times 168 &= 2352 \equiv 180 \equiv -1 \pmod{181} \\
14^{45} &\equiv 1 \pmod{181} \quad (\text{power witness}) \\
14^{15} &\equiv 48 \neq 1, \quad 14^9 \equiv 135 \neq 1 \pmod{181} \quad (\text{non-identities for } 3, 5) \\
168^{90} &\equiv 1 \pmod{181} \quad (\text{power witness}) \\
168^{45} &\equiv 180 \neq 1, \quad 168^{30} \equiv 48 \neq 1, \quad 168^{18} \equiv 42 \neq 1 \pmod{181} \\
\omega_{181} &= 168^{30} \equiv 48, \quad 48^3 \equiv 1, \quad 1 + 48 + 48^2 \equiv 0 \pmod{181}
\end{aligned}$$

A.2 La Rue's Vertex ($p = 229$)

$$\begin{aligned}
148^2 - 148 - 1 &= 21755 \equiv 0 \pmod{229} \quad (\varphi = 148) \\
82^2 - 82 - 1 &= 6641 \equiv 0 \pmod{229} \quad (\bar{\varphi} = 82) \\
148 \times 82 &= 12136 \equiv 228 \equiv -1 \pmod{229} \\
148^{114} &\equiv 1 \pmod{229} \quad (\text{power witness}) \\
148^{57} &\equiv 228 \neq 1, \quad 148^{38} \equiv 94 \neq 1, \quad 148^6 \equiv 44 \neq 1 \pmod{229} \quad (\text{non-id. for } 2, 3, 19) \\
82^{57} &\equiv 1 \pmod{229} \quad (\text{power witness}) \\
82^{19} &\equiv 94 \neq 1, \quad 82^3 \equiv 165 \neq 1 \pmod{229} \quad (\text{non-identities for } 3, 19) \\
\omega &= 82^{19} \equiv 94, \quad 94^2 \equiv 134, \quad 94^3 \equiv 1, \quad 1 + 94 + 134 = 229 \equiv 0 \pmod{229}
\end{aligned}$$

A.3 Mirror Vertex and Lockwood Decomposition

$$\begin{aligned}
14^{15} \bmod 181 &= 48 = 229 \bmod 181 \quad (229 \text{ in golden orbit}) \\
168^{63} \bmod 181 &= 139 \quad (139 \text{ in conjugate orbit}) \\
14^{41} \bmod 181 &= 144 \quad (\text{Lockwood side: chromatic}) \\
168^{55} \bmod 181 &= 138 \quad (\text{Lockwood side: chromomatic}) \\
168^{25} \bmod 181 &= 116 \quad (\text{Lockwood side: chromomatic})
\end{aligned}$$

A.4 Cyber Prismatic Formulas

Formula 1 (Color Projector):

$$\begin{aligned} 82^{19} \bmod 229 &= 94 \quad (\bar{\varphi}^{19} = \omega) \\ 94^3 \bmod 229 &= 1, \quad (1 + 94 + 134) \bmod 229 = 0 \\ 57/19 &= 3 \quad (\text{color rank}) \end{aligned}$$

Formula 2 (BitCollapse = Bridge Balance):

$$148 \times 82 = 12136 \equiv 228 \equiv -1 \pmod{229} \quad (\varphi \cdot \bar{\varphi} = -1)$$

Formula 4 (Translation Phase Shift):

$$\begin{aligned} 148^9 \bmod 229 &= 15 \quad (\text{multiplier} = \varphi^9) \\ 82^{15} \bmod 229 &= 17, \quad 148^{51} \bmod 229 = 26 \\ 15 \times 17 &= 255 \equiv 26 \pmod{229} \quad (\text{orbit crossing}) \end{aligned}$$

Formula 5 (Gauge Arc Rotation):

$$94 \times 82 \bmod 229 = 82^{20} \bmod 229 \quad (\omega \cdot \bar{\varphi} = \bar{\varphi}^{20})$$

A.5 Python Verification Script

```
# Digital Wave Mechanics -- complete verification
# Run: python3 verify_digital_wave.py (53+ checks, zero deps)
# Available at: github.com/Dielawn-01/ProtorealZeta/papers/

# Lockwood's vertex:
assert pow(14,45,181) == 1          # ord(phi)=45
assert pow(168,45,181) == 180      # phibar^45 = -1, NOT 1

# Cyber Prismatic formulas:
assert pow(82,19,229) == 94        # CP1: color projector
assert (1+94+pow(94,2,229))%229==0 # confinement
assert pow(148,9,229) == 15        # CP4: translation key
assert (15*17)%229 == 26           # CP4: orbit crossing

# D-Neutrino / Infoton:
import math
kB, T = 1.380649e-23, 300
E_info = kB * T * math.log(2)      # Walker infoton
m_nu = 0.8                         # KATRIN bound (eV)
eV = 1.602176634e-19
assert abs(m_nu*eV/E_info - 44.64) < 0.01 # ~45
assert 57 // 19 == 3               # 3 neutrino flavors
```

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